Principles of inheritance and Variation

Introduction and Mendel's Law of Inheritance

1. The process by which characters are passed on from parent to progeny is called
1) Transformation
2) Independent assortment
3) Inheritance
4) All of the above

2. An inherited character and its detectable variant is called
1) Factor
2) Gene
3) Trait
4) Allele

3. Select the correct option for the words to fill in the blanks A, B and C:
(i) A are the unit of inheritance.
(ii) Tall and dwarf represent contrasting form of a B called height.
(iii) C are slightly different form of the same gene.

	A	B	C
1)	Genes	Trait	Alleles
2)	Genes	Character	Traits
3)	Factors	Gene	Alleles
4)	Factors	Gene	Traits

4. How many of the following trait(s) in Pisum sativum is/are dominant?

Dwarf stem height, Violet flower colour, Axial flower position, Yellow pod colour, Inflated pod shape, Round seed shape

1) Four
2) Three
3) Two
4) Five

5. Which of the following is the correct dominant recessive pair of pea plant?
1) Pod shape - constricted (dominant) and full (recessive)
2) Pod colour - green (dominant) and yellow(recessive)
3) Flower position - terminal (dominant) and axial (recessive)
4) Seed shape - round (recessive) and wrinkled (dominant)

6. Read the following statements and choose the correct option.
I. Variation is the degree by which progeny differ from their parents.
II. Humans exploited the variations that were naturally present in the wild populations of plants and animals to selectively breed and select for organisms that possessed desirable characters.
III. Though our ancestors knew about the inheritance of characters and variations, they had very little idea about the scientific basis of these phenomena.

Correct statement(s) is/are
1) Only II and III
2) Only II
3) Only I and II
4) All of the above

7. Number of contrasting trait(s) related to flower of pea plants studied by Mendel is

1) Two
2) Four
3) Seven
4) Three

8. A true breeding line is one that
1) Has undergone continuous self-pollination
2) Has undergone continuous cross pollination
3) Shows stable trait inheritance and expression for several generations
4) Both (1) and (3)

9. Organisms phenotypically similar but genotypically different are said to be
1) Homozygous
2) Multizygous
3) Heterozygous
4) Hemizygous

Inheritance of One Gene

10. Which of the following cannot be explained on the basis of Mendel's law of dominance?
1) The discrete unit controlling a particular character is called a factor.
2) Out of one pair of factors, one is dominant and the other is recessive.
3) Factors occur in pairs.
4) Alleles do not show any blending and both the characters recover as such in F, generation.

11. In hybridization, Tt x tt gives rise to the progeny of ratio
1) 1:1
2) 1:2
3) 2:1
4) 1:2:1

12. The graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross is known as
1) Back cross
2) Pedigree
3) Punnett square
4) Central dogma

13. Read the given statements of assertion and reason and select the correct option.
Assertion: Dominance is not an autonomous feature of a gene or the product that it has information for.
Reason: Dominance depends much on the gene product and the production of a particular phenotype from this product.
1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

14. The genotypic and phenotypic ratio in a monohybrid cross is, respectively
1) 3:1,1:2:1
2) 1:2:1,3:1
3) 3: 1,1:1
4) 1: 1,1:1

15. The law based on the fact that the alleles do not show blending also proposes that

(A) A gamete receives only one of the two factors.
(B) Both the characters are recovered as such in F_2 generation.
1) Both (A) and (B) are incorrect
2) (A) is incorrect but (B) is correct
3) Both (A) and (B) are correct
4) (B) is incorrect but (A) is correct

16. Match the column I with column II and choose the correct option.

Column-I		Column-II	
a	Incomplete dominance	1	ABO blood grouping
b	Co-dominance	2	Snapdragon
c	Multiple allele	3	Garden peas
d	Dominance	4	Fruit-flies

1) a-2, b-4, c-3, d-1
2) a-1, b-2, c-4, d-3
3) a-3, b-4, c-1, d-2
4) a-2, b—1, c-1, d-3

17. A phenomenon in which neither of the alleles of a gene is completely dominant over the other and the F_1 individual is intermediate between the two parents is called
1) Incomplete dominance
2) Complete dominance
3) Codominance
4) Epistasis

18. 'Each gamete is pure for a trait', given statement explains
1) Law of dominance
2) Law of segregation
3) Law of independent assortment
4) Linkage

19. Read the given statements and select the correct option.
Statement I: In snapdragon, F_1 plants do not have red or white flowers.
Statement II: It is an intermediate inheritance with neither of the two alleles of a gene being dominant over each other.
1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true, and statement II is false.
4) Statement I is false, and statement II is true.

20. For the ABO blood group system of the human, which is correct?
(i) Multiple allele
(ii) Incomplete dominance
(iii) Co-dominance
(iv) Dominance
(v) Polygenic inheritance
(vi) Pleiotropy
1) i, ii, v, vi
2) ii, iv, vi
3) i, iii, iv, vi
4) i, iii, iv

21. Person having genotype $I^A I^B$ would show the blood group as AB. This is because of
1) Pleiotropy

2) Co-dominance
3) Segregation
4) Incomplete dominance

22. If a cross made between two individuals produces offspring with 50% dominant character (A) and 50% recessive character (a), then the genotypes of the parents are
1) Aa x Aa
2) Aa x aa
3) AA × aa
4) AA × aa

23. Genotype of dominant trait, which may be either homozygous or heterozygous, can be determined by
1) Reciprocal cross
2) Outcross
3) Test cross
4) Pedigree analysis

24. Read the following statements and state true (T) and false (F):
A. The genotypic ratio of flower colour in snapdragon is exactly similar to genotypic ratio of Mendelian monohybrid cross.
B. In a diploid organism, there are four copies of each gene.
C. If we want to determine the genotype of violet flowered pea plant, then it is crossed with recessive parent.

	A	B	C
1)	T	F	T
2)	T	T	T
3)	F	F	T
4)	F	T	T

25. Codominance is different from incomplete dominance as in codominance
1) The hybrid is intermediate
2) Only recessive allele is expressed in F_1
3) Both the alleles are expressed equally
4) Genotypic ratio is 1:1

26. Which of the following is incorrect about Punnett square?
1) The possible gametes are written on two sides, usually the top row and left columns.
2) It is a graphical representation to calculate the probability of all possible phenotypes of offspring in a genetic cross.
3) All possible combinations are represented in boxes below in the squares, which generates a square output form
4) None of the above

27. Match the following columns and choose the correct option.

Column-I		Column-II	
1	Dominance	a	F_1 generation resembles both parents
2	Incomplete dominance	b	F_1 resembles either of the two parents
3	Co-dominance	c	F_1 generation is in between two parents

1) 1-b, 2-a, 3-c
2) 1-c, 2-b, 3-a
3) 1-a, 2-c, 3-b
4) 1-b, 2-c, 3-a

28. Read the following statement having two blanks A and B: The functioning allele which represents the original phenotype is the A allele and the modified allele is generally B allele.

	A	B
1)	Dominant	Recessive
2)	Recessive	Dominant
3)	Dominant	Incomplete dominant
4)	Recessive	Codominant

29. Which of the following was not the observation made by Mendel in his experiment with garden peas?
1) Every character is controlled by discrete units called factors.
2) The factors occur in pair.
3) In a dissimilar pair of factors, only one is able to express, called recessive factors.
4) Both factors of a pair segregate from each other during gamete formation.

Inheritance of Two Genes

30. In a typical Mendelian cross, which is a dihybrid cross, one parent is a homozygous for both dominant traits and another parent is homozygous for both recessive traits. In F_2 generation, both parental combinations and recombinants appear. The phenotypic ratio of parental combinations to recombinants is
1) 10:6
2) 9:7
3) 15:1
4) 12:4

31. What is the ratio of round green and wrinkled yellow seeds in F_2 generation of Mendelian dihybrid cross?
1) 9:1
2) 1:1
3) 1:3
4) 1:3

32. In a dihybrid cross, if you get 9:3:3: 1 ratio, it denotes that
1) The alleles of two genes are interacting with each other
2) It is a multigenic inheritance
3) It is a case of multiple allelism
4) The alleles of two genes are segregating independently

33. Frequency of pure round seeded pea plants in F_2 generation of a typical dihybrid cross is
1) 12.5%
2) 25%
3) 6.25%
4) 56.25%

34. Law of independent assortment can be explained by
1) Monohybrid cross
2) Reciprocal cross
3) Dihybrid cross
4) Test cross

35. Read the following statements regarding dihybrid cross (RrYy) and choose the correct option.

I. In segregation of one pair of genes R and r. Fifty per cent of the gametes have the gene R and the other 50 per cent have r.

II. Besides each gamete having either R or r, it should also have the allele Y or y.

III. There are four genotypes of gametes (four types of pollen and four types of eggs), i.e., RY, Ry, rY and ry each with a frequency of 50 per cent or 1/2 of the total gametes produced.

How many statement(s) is/are incorrect?

1) Three
2) Two
3) One
4) None of the above

36. All were responsible for rediscovery of Mendel's result of inheritance of characters, except

1) de Vries
2) Correns
3) Morgan
4) Tschermak

37. Morgan coined the term recombination to describe

1) Physical association of geneson a chromosome
2) Generation of new traits with discontinuous variations
3) Generation of non-parental gene combinations
4) Linkage of two genes on different chromosomes

38. Experimental verification of the chromosomal theory of inheritance was done by

1) Sutton
2) T. H. Morgan
3) Boveri
4) Mendel

39. The recombination frequency between the genes a and b is 5%, a and d is 10%, b and d is 15%, c and d is 8%, b and c is 23%. What will be the sequence of these genes on a linear chromosome?

1) b, a, d, c
2) a, c, b, d
3) b, d, a, c
4) c, a, d, b

40. Linkage and crossing over are

1) Inversely proportional to each other
2) Directly proportional to each other
3) Not related to each other
4) Same phenomenon

41. Read the given statements of assertion and reason and select the correct option.

Assertion: When two genes in a dihybrid cross were situated on the same chromosome, the proportion of parental gene combinations were much higher than the non-parental type.

Reason: This is due to the physical association or linkage of the two genes.

1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

42. Drosophila melanogaster was found to be suitable for experiments by T. H. Morgan due to some of its specific features. These features are all, except

1) Clear differentiation of sexes
2) Easy growth on simple synthetic medium in the laboratory
3) They complete their life cycle within two weeks

4) Single mating produces small number of progeny flies

Polygenic Inheritance and Pleiotropy

43. Polygenic inheritance

A. Traits are controlled by three or more genes.

B. Takes into account the influence of environment

C. The phenotype reflects the contribution of each allele and the effect of each dominant allele is additive.

1) Only (A) is correct
2) Only (B) is correct
3) Only (B) and (C) are correct
4) All (A), (B) and (C) are correct

44. Which of the following is/are an example of pleiotropy?

1) Haemophilia
2) Phenylketonuria
3) Turner's Syndrome
4) All of the above

45. Occasionally, a single gene may express more than one effect. The phenomenon is called

1) Multiple allelism
2) Mosaicism
3) Pleiotropy
4) Polygenic inheritance

46. Read the following statements about polygenic inheritance and choose the correct option.

I. In a polygenic trait, the phenotype reflects the contribution of each allele, i.e., the effect of each allele is additive.

II. Besides the involvement of multiple genes, polygenic inheritance also takes into account the influence of environment.

III. The number of each type of alleles in the genotype would determine the darkness or lightness of the skin in an individual.

IV. The dominant forms A, B, and C are responsible for the dark skin colour and the recessive forms a, b, and c for light skin colour.

Correct statement(s) is/are

1) Only II
2) Only I and III
3) Only II and IV
4) All of the above

47. If skin colour in humans is controlled by three gene pairs, then what is the number of intermediates and parental combinations in F_2 generation?

1) 2, 62
2) 62, 2
3) 2, 14
4) 14,2

48. A single gene product may produce more than one effect in

1) Criss cross inheritance
2) Pleiotropy
3) Polygenic inheritance
4) Quantitative inheritance

49. Which of the following show continuous variation in a population?
1) Human skin colour
2) Cob length in maize
3) Kernel colour in wheat
4) All of these

50. Select the incorrect statement w.r.t. polygenic inheritance.
1) The dominant alleles have cumulative effect.
2) Intermediates are more frequent.
3) A single gene exhibits multiple phenotypic expression
4) None of the above

51. Three genes A, B, C control skin colour in human. The genotypewith all the recessive alleles (aabbcc) will have the
1) Dark skin colour
2) Wheatish skin colour
3) Lightest skin colour
4) Intermediate skin colour

52. Phenylketonuria manifests itself through phenotypic expression characterised by
i. Reduction in hair pigmentation
ii. Pattern baldness
iii. Mental retardation
iv. Cola coloured urine
v. Reduction in skin pigmentation
1) Only ii and iv
2) Only i, iii and v
3) Only ii, iii and v
4) i, ii, iii and iv

Sex Determination and Mutation

53. Read the following statements:
A. This phenomenon results in alteration of DNA sequence and consequently results in changes in genotype and the phenotype of an organism.
B. In addition to recombination, it is another phenomenon that leads to variation in DNA.
The phenomenon explained above is
1) Linkage
2) Coupling and repulsion of genes
3) Mutation
4) Fertilization

54. Fill in the blanks and select the correct option:
(i) A could trace a specific nuclear structure, all through spermatogenesis in a few insects.
(ii) Grasshopper is an example of B type of sex determination.
(iii) ale bird is C
(iv) Rest of chromosomes, except sex chromosomes, are known as D

	A	B	C	D
1)	Henking	XO	Heterogametic	Autosomes
2)	Morgan	XY	Heterogametic	Autosomes
3)	Hugo de Vries	XO	Heterogametic	Autosomes
4)	Henking	XO	Heterogametic	Autosomes

55. Read the following statements and select the correct option:

Statement A: In humans, sex determination mechanism is designated to be the example of male heterogamety.

Statement B: In humans, sex of the child is determined by mother and not by father.

1) Only (A) is incorrect
2) Only (B) is incorrect
3) Both (A) and (B) are incorrect
4) Both (A) and (B) are correct

56. Match column I with column II and choose the correct option

Column-I		Column-II	
a	Chromosomal aberration	1	UV radiations
b	Polyploidy	2	Cancer cells
c	Mutagen	3	Sickle cell anaemia
d	Point mutation	4	Often seen in plants

1) a-2, b-4, c-3, d—1
2) a-3, b—1, c—2, d-4
3) a-2, b-4, c-1, d-3
4) a-4, b-3, c—2, d—1

57. In which of the following, eggs fertilised by spermhaving an X-chromosome become female and, those fertilised by sperms not having an X-chromosome become males?

1) Grasshopper
2) Humans
3) Birds
4) All of these

58. The AA + ZW type of chromosome condition is seen in

1) Male bird
2) Female bird
3) Male grasshopper
4) Male Drosophila

59. The probability of either a male or a female child in each pregnancy is always

1) ½
2) 3/2
3) ¾
4) 1/4

60. Chromosomal aberrations are commonly observed in _____

1) Cancer cells
2) Tapetum cells
3) Meristematic cells
4) Quiescent centre of roots

61. Polyploidy is

A. The occurrence of more than two sets of chromosomes.
B. A result of failure of cytokinesis after telophase stage.
C. More common in plants.

1) Only (A)
2) Only (A) and (C)
3) Only (A) and (B)
4) (A), (B) and (C)

62. In which of the following pair of chromosomal aberrations, both result in change in gene number?

1) Inversion, duplication
2) Duplication, deletion
3) Duplication, translocation
4) Inversion, translocation

63. Sex determination in grasshopper, humans, and Drosophila is similar because

1) Females are hemizygous
2) Male have one X-chromosome and female have two X-chromosomes
3) Females shows heterogamety
4) The ratio of autosomes and sex chromosomes is the same in all three organisms

64. Read the following statements and state true (T) and false (F):

A. In male grasshoppers, 50% of the sperm has no autosomes.
B. Usually female birds produce two types of gametes based on sex chromosomes.
C. Failure of segregation of chromatids during cell division results in the gain or loss of chromosome.
D. Henking could trace a specific nuclear structure all through oogenesis of few insects.

	A	B	C	D
1)	T	F	F	T
2)	F	T	T	F
3)	T	T	F	F
4)	F	T	F	T

65. ZZ/ZW type of sex determination is seen in

1) Platypus
2) Snails
3) Cockroach
4) Peacock

66. Match column I with column II and choose the correct combination from the options given below.

Column-I		Column-II	
a	Chromosomal disorder	1	Drosophila
b	XO type sex determination	2	Grasshopper
c	XY type sex determination	3	Birds
d	ZW type sex determination	4	Klinefelter's syndrome

	a	b	c	d
1)	1	2	3	4
2)	4	3	2	1
3)	4	2	1	3
4)	2	1	4	3

67. Read the following statements and select the correct option:

A. In honeybee, males do not have father and thus cannot have sons but have a grandfather and can have grandsons.

B. In birds, sperm determines the sex of the offspring.

1) Only (A) is correct
2) Only (B) is correct
3) Both (A) and (B) are correct
4) Both (A) and (B) are incorrect

68. Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome(s) is called
1) Polyploidy or transition
2) Point mutation or euploidy
3) Hypoploidy or hyperploidy
4) Hyperploidy or transversion

69. Sex-determination mechanism in honeybee is
1) ZZ-ZW type
2) XX-XO type
3) XX-XY type
4) Haploid-diploid mechanism

70. In a certain taxon of insects, some have 17 chromosomes and the others have 18 chromosomes. The 17 and 18 chromosome-bearing organisms are
1) Males and females, respectively
2) Females and males, respectively
3) All males
4) All females

71. Read the following statements and choose the incorrect option.
1) One DNA helix runs continuously from one end to the other in each chromatid, in a highly supercoiled form
2) Loss (deletions/duplication) or gain (insertion) of a segment of DNA, result in alteration in chromosomes.
3) There are many chemical and physical factors that induce mutations.
4) Since genes are known to be located on chromosomes, alteration in chromosomes results in abnormalities or aberrations.

72. Statement I: In birds, the chromosome composition of the egg determines the sex.
Statement II: Female birds are heterogametic.
1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true, and statement II is false.
4) Statement I is false, and statement II is true.

Genetic Disorders

73. Mark the correct statements:
I. There are many chemical and physical factors that induce mutations.
II. Pedigree study provides a strong tool, which is utilized to trace the inheritance of specific trait, abnormality or disease.
III. The pattern of inheritance can be traced in a family by the pedigree analysis.
1) All are correct
2) (I) and (II) are correct
3) (II) and (III) are correct
4) (I) and (III) are correct

74. What percentage of male (A) and female (B) get affected by colour blindness in humans?

	A	**B**
1)	8%	0.4%
2)	0.4%	8%
3)	50%	50%

4)	8%	20%

75. Of a normal couple, half of the sons are haemophilic, and half of the daughters are carrier, the genes for this disease in the couple are located on

1) X-chromosome of both parents
2) Only the X-chromosome of father
3) Only one X-chromosome of mother
4) Both the X-chromosome of mother

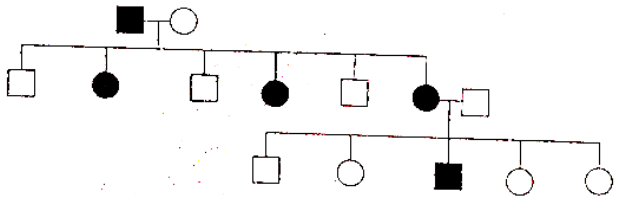
76. In a-thalassemia

(A) α -globin chain is produced in less number.
(B) Two closely linked genes HBA1 and HBA2 on chromosome 16 are affected.
(C) Incorrectly functioning globin chain is synthesized.

Correct statements is/are

1) Only (A)
2) Only (B)
3) Only (A) and (B)
4) Only (B) and (C)

77. Study the given Pedigree and identify the given trait:



1) utosomal dominant
2) Autosomal recessive
3) X-linked dominant
4) X-linked recessive

78. All are autosomal recessive disorders, except

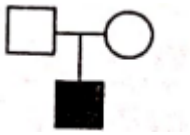
1) Cystic fibrosis
2) Thalassemia
3) Phenylketonuria
4) Haemophilia

79. Study the two cases carefully:

	Mother	Father	Children
Case I	Normal	With disease	Daughters always with disease
Case II	With disease	Normal	Sons and daughters could show disease

	Case I	**Case II**
1)	X-linked dominant disease	Autosomal recessive disease
2)	Autosomal recessive disease	X-linked recessive disease
3)	X-linked recessive disease	Autosomal dominant disease
4)	Autosomal dominant disease	X-linked dominant disease

80. Which of the following is correct for the given pedigree?



1) Disease present in child may be X-linked dominant.
2) Considered trait is autosomal recessive.
3) Considered trait is Y-linked.
4) None of the above

81. An individual shows the following characteristics:

A. Sterile females

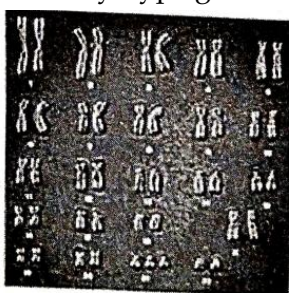
B. Rudimentary ovaries

C. Lack of secondary sexual characters

The individual suffers from

1) Klinefelter's syndrome
2) Turner's syndrome
3) Edward's syndrome
4) Down's syndrome

82. The karyotype given below is related to



1) Turner's syndrome
2) Down's syndrome
3) Klinefelter syndrome
4) Edward's syndrome

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	3	1	1	2	4	2	4	3	4	1	3	1	2	3
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
4	1	2	1	4	2	2	3	1	3	2	4	1	3	1
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
2	4	2	3	3	3	3	2	1	1	1	4	4	2	3
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
4	2	2	4	3	3	2	3	4	2	3	1	2	1	1
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
4	2	2	2	4	3	1	3	4	1	2	1	1	1	3
76	77	78	79	80	81	82								
3	3	4	1	2	2	2								